Probability Review

Probability space $(\Omega, \mathcal{A}, P)$ models real world experiment with random outcome

- Ω: sample space — set of all possible outcome
- $\mathcal{A}$: event space — potential results of an experiment
- P: probability function $P(A)$

Example toss a coin 2 times

$\Omega = \{hh, ht, th, tt\}$

$(A \in \mathcal{A})$ $A \subseteq \Omega$ describes a particular outcome subset

event: two tosses with same outcome

event $A = \{hh, tt\} \subseteq \Omega$

$P: \mathcal{A} \to [0,1]$

$P(A)$ = prob. of A occuring

$P(\Omega) = 1$

Exercise, represent A as a R.V. y.

$T = \{0,1\}$

$y: \Omega \to \{0,1\}$

$y = 1\{a_1 = a_2\}$

where a_1, a_2 are two tosses

Random Variables (R.V.) $X: \Omega \to T$

T: target space } discrete set, $\mathbb{R}, \mathbb{R}^n$

example let X be the # of heads in 2 tosses.

$w \in \Omega$	hh	ht	th	tt
X				
$X(w)$	2	1	1	0

$T = \{0,1,2\}$

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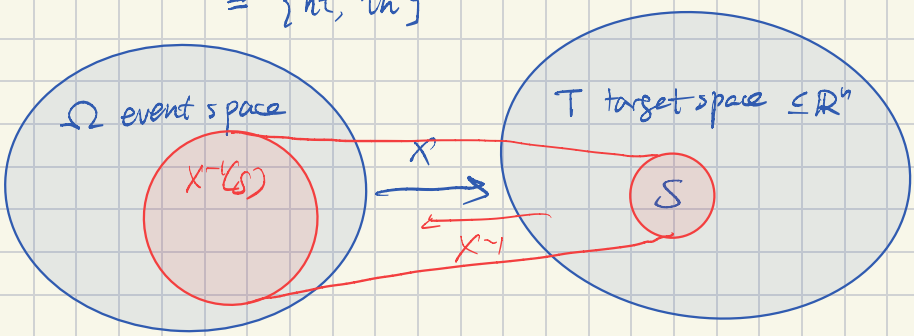
For any $S \subseteq T$, we can define $P_X(S) \in [0,1]$ as the prob. to some event occuring corresponding to R.V. X.

e.g. let $S = \{1\}$ (exact one head in 2 tosses) $\quad X : \Omega \to \{0,1,2\}$ (T)

$\{hh, ht, th, tt\}$

$X^{-1}(S) = \{w \in \Omega : X(w) \in S\} \subseteq \Omega$

$= \{ht, th\}$



For $S \subseteq T$,

$P_X(S) = P(X \in S) = P(X^{-1}(S)) = P(\{w \in \Omega \mid X(w) \in S\})$

prob. in target space — prob. in event space

we're interested in

X is distributed according to Prob. P_X.

When T is

1) finite or countable infinite) → X is a discrete R.V.

e.g $\{0,1,2\}$ $\quad \mathbb{N}$

2) infinite → X is a continuous R.V.

e.g $\mathbb{R}$, $\mathbb{R}^n$

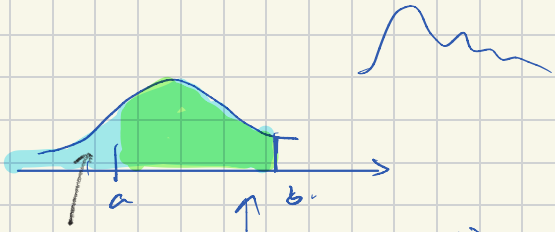
Discrete R.V. X, probability mass function $P(X=x)$ $\quad \sum_{x \in \Omega} P_X(x) = 1$

Continuous R.V. ($T = \mathbb{R}$) Given $a, b \in \mathbb{R}, a < b$.

"PDF" probability density function $\quad P(a \le X \le b)$

"CDF" cumulative probability function $\quad P(X \le x)$

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CDF: $F_X(x) = Pr(X \le x)$ PDF: $Pr(a \le X \le b)$

[def] A function $f: \mathbb{R}^D \to \mathbb{R}$ is a probabilistic density function (pdf) if

① $\forall x \in \mathbb{R}^d$, $f(x) \ge 0$ (non-negative)

② its integral exists and $\int_{\mathbb{R}^d} f(x)\,dx = 1$

$P(a \le X \le b) = \int_a^b f(x)\,dx$, $a, b \in \mathbb{R}$, $x \in \mathbb{R}$.

[def] Multivariate R.V. X, (x value) $\in \mathbb{R}^d$.

cumulative distribution function (cdf)

$F_X(x) = P(X_1 \le x_1, \cdots, X_D \le x_D)$

(X: R.V.; joint prob.)

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_D \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix}$$

$$F_X(x) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_D} f(z_1, \cdots, z_D)\,dz_1 \cdots dz_D$$

convention

R.V. → X, Y

values → x, y

$P_r(X=x)$

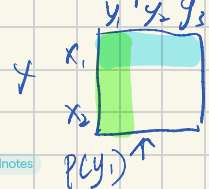
$P_X(x) \xrightarrow{\text{shorten}} P(x)$

Sum rule

1) y is discrete, $\sum_{y \in Y} P(x,y) = P(x)$

2) y is continuous $\int_y P(x,y)\,dy$

} marginalization



$\leftarrow \sum_{i=1}^{3} p(x_1, y_i) = P(x_1)$

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product rule

$$p(x,y) = p(y|x)\,p(x)$$
$$= p(x|y)\,p(y)$$

marginal likelihood

$$p(y) = \int p(y|x)\,p(x)\,dx = \mathbb{E}_x[p(y|x)]$$

$p(y)$: prior

def Expected value of function $g: \mathbb{R} \to \mathbb{R}$ of a random var. $X \sim p(x)$ is

$$\mathbb{E}_x[g(x)] = \int_X g(x)\,p(x)\,dx \quad \text{(continuous } X\text{)}$$

$$\mathbb{E}_x[g(x)] = \sum_{x \in X} g(x)\,p(x) \quad \text{(discrete } X\text{)}$$

↝ generalize to $X \in \mathbb{R}^D$

↑ main summary statistics of $p(x)$

def mean of a R.V. $X \in \mathbb{R}^D$ is the an average defined by

$$\mathbb{E}_X[x] = \begin{bmatrix} \mathbb{E}_{X_1}[x_1] \\ \vdots \\ \mathbb{E}_{X_D}[x_D] \end{bmatrix} \in \mathbb{R}^D$$

($g = 1$)

where $\mathbb{E}_{X_d}[x_d] = \int_X x_d\,p(x_d)\,dx_d$

$d = 1, \ldots, D$

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proposition For any $f(x) = ag(x) + bh(x)$, $a, b \in \mathbb{R}$, $x \in \mathbb{R}^D$

$$\mathbb{E}_x[f(x)] = a\mathbb{E}_x[g(x)] + b\mathbb{E}_x[h(x)]$$

proof

$$\mathbb{E}_x[ag(x) + bh(x)] = \int [ag(x) + bh(x)] p(x) dx$$

$$= a\int g(x)p(x)dx + b\int h(x)p(x)dx$$

$$= a\mathbb{E}[g(x)] + b\mathbb{E}[h(x)]$$

def Covariance of R.V. $X, Y \in \mathbb{R}$

$$\mathrm{cov}_{X,Y}[x, y] = \mathbb{E}_{XY}[(x - \underbrace{\mathbb{E}_x[x]}_{\text{mean } \mu_x})(y - \underbrace{\mathbb{E}_y[y]}_{\text{mean } \mu_y})]$$

def

$$\mathrm{cov}_{X,Y}[x, y] = \mathbb{E}[xy] - \mathbb{E}[x]\mathbb{E}[y]$$

$X \in \mathbb{R}^{D_1}$, $y \in \mathbb{R}^{D_2}$ $\quad$ $(D_1 \times 1)$ $(1 \times D_2)$

$$\mathrm{Cov}[x, y] = \mathbb{E}[xy^T] - \mathbb{E}[x]\mathbb{E}[y]^T$$

$$= \mathrm{Cov}(y, x) \in \mathbb{R}^{D_1 \times D_2}$$

def variance $\mathrm{var}(x) = \mathrm{cov}_x[x, x]$

(covariance matrix). $\mathrm{Cov}_x[x, x] = \mathbb{E}_x[(x-\mu)(x-\mu)^T]$

$$= \mathbb{E}_x[xx^T] - \mathbb{E}_x[x]\mathbb{E}_x[x]^T$$

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$$\text{Cov}(X) = \text{Cov}_X[X, X] = \begin{bmatrix} \text{Cov}[X_1, X_1] = \text{Var}(X_1) & \text{Cov}[X_1, X_2] & \cdots & \text{Cov}(X_1, X_D) \\ & \text{Var}(X_2) & & \\ \vdots & & \ddots & \\ \text{Cov}(X_D, X_1) & & \cdots & \text{Var}(X_D) \end{bmatrix}$$

Correlation

$$\text{Corr}(x, y) = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x)}\sqrt{\text{Var}(y)}}$$

$\sqrt{\text{Var}(x)} \to \text{std}(x)$, $\sqrt{\text{Var}(y)} \to \text{std}(y)$

$\text{Cov} \to b^2$

$\text{std} \to b$.

$$\text{Cov}\left(\frac{x}{\text{std}(x)}, \frac{y}{\text{std}(y)}\right)$$

Empirical mean / covariances

Given $x^{(1)}, \ldots, x^{(N)} \in \mathbb{R}^D$.

empirical mean $\bar{x} := \frac{1}{N}\sum_{i=1}^{N} x^{(i)}$ $(\hat{\mu})$ $\hat{\mathbb{E}}_D[x]$

empirical covariance

$$\Sigma := \frac{1}{N}\sum_{n=1}^{N}(x^{(i)} - \bar{x})(x^{(i)} - \bar{x})^T$$

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